

Assignment 7

Your Name Here

A microbe called *Cinderspore* is spreading between star systems and dimming stars as it goes. One research vessel, the *Meridian*, was sent out with a small crew to study it and find a way to stop it before it reaches Earth.

The *Meridian*'s crew is busy keeping the ship running and the mission alive while you are analyzing the data sent by Frank. We have a new transmission:

Mission Log 007 — from Frank

Analyst! Hi! We have spent a lot of time asking whether groups are different. Command has moved on to asking how things change *together*. Does more sleep go with faster reflexes? Can we predict Cinderspore growth from temperature? And can we explain scrubber output using more than one thing at a time? I said yes to that last one before checking with you. I hope that was a reasonable thing to have said.

Question 0: Data Import

0a. Call in any packages that are needed for this assignment.

0b. Import this week's datasets here.

Question 1: Sleep and Reflexes

Frank: "We logged sleep hours and reflex-drill times for 50 crew members across the fleet. I think the people getting more sleep respond faster, but I would like something better than 'Frank looked at the spreadsheet and nodded.'"

This question will use the `crew_reflex` dataset. Each row represents a different crew member.

Variable	Type	Description
<code>sleep_hours</code>	Continuous	Hours slept in the rest cycle before the drill
<code>reflex_time_ms</code>	Continuous	Time to respond during the reflex drill (milliseconds)

1a. Construct a scatterplot with `sleep_hours` on the x axis and `reflex_time_ms` on the y axis.

1b. Describe the direction, form, and strength of the relationship. Identify any apparent outliers.

Your answer here.

1c. Calculate Pearson's correlation between sleep hours and reflex time. Report the coefficient and interpret its direction and strength in context.

Your answer here.

1d. Calculate Spearman's correlation between sleep hours and reflex time. Report the coefficient and interpret its direction and strength in context.

Your answer here.

1e. Construct the appropriate plot to assess the assumption for examining Pearson's correlation.

1f. Which correlation coefficient is appropriate to report? Why?

Your answer here.

1g. Perform the appropriate two-sided hypothesis test to determine if there is an association between sleep hours and reflex time.

Hypotheses:

- H_0 :
- H_1 :

Test Statistic and p -value

- *statistic* =
- p =

Rejection Region

- Reject H_0 if ...

Conclusion and Interpretation

- Reject or fail to reject H_0 .
- There is or is not sufficient evidence to conclude...

1h. Frank says, "So this proves that making everyone sleep longer will make their reflexes faster!" Does this analysis justify his claim? Explain.

Your answer here.

Question 2: Predicting Cinderspore Growth

Frank: "Science ran 40 separate culture trials, each at a different recorded temperature. Command wants to know whether we can use temperature to predict how quickly Cinderspore grows. I would personally prefer it grew at zero spores per hour, but apparently that's not a realistic model."

This question will use the `culture_growth` dataset. Each row represents a separate culture trial. The observed temperatures range from 18°C to 30°C.

Variable	Type	Description
temperature_c	Continuous	Temperature maintained during the trial (°C)
growth_rate	Continuous	Estimated Cinderspore growth rate (spores/hr)

2a. Identify the response variable and the explanatory variable.

Your answer here.

2b. Construct a scatterplot of growth rate against temperature. Does a straight line appear to be a reasonable way to describe the relationship? Explain.

Your answer here.

2c. Fit a simple linear regression model predicting growth_rate from temperature_c. Display the model results.

2d. State the model constructed.

$$\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x_1 + \cdots + \hat{\beta}_k x_k$$

2e. Construct a QQ plot of the residuals and a plot of residuals against fitted values. Assess normality, linearity, and constant variance. Justify your answers using the graphs.

Your answer here.

2f. Interpret the estimated slope in context, including units.

Your answer here.

2g. Construct the hypothesis test for the slope to determine if there is a linear relationship between temperature and Cinderspore growth rate.

Hypotheses:

- H_0 :
- H_1 :

Test Statistic and p -value

- $statistic =$
- $p =$

Rejection Region

- Reject H_0 if ...

Conclusion and Interpretation

- Reject or fail to reject H_0 .
- There is or is not sufficient evidence to conclude...

2h. Frank says, “So if we turn down the temperature, Cinderspore will definitely grow more slowly in every culture!” Does this analysis justify his claim? Explain what the model tells us about average growth and what it can predict for an individual culture.

Your answer here.

Question 3: What Explains Scrubber Output?

Frank: “Engineering sent 125 independent scrubber test runs. We logged airflow, filter age, humidity, inlet temperature, and vibration. Command wants to know which measurements are related to oxygen output when we account for the others. I logged five things. That does not mean all five deserve credit.”

This question will use the `scrubber_output` dataset. Each row represents a separate test run.

Variable	Type	Description
<code>o2_output_lpm</code>	Continuous	Oxygen output (liters/minute)
<code>airflow_lpm</code>	Continuous	Airflow entering the scrubber (liters/minute)
<code>filter_age_hours</code>	Continuous	Hours of use on the filter at the start of the run
<code>humidity_pct</code>	Continuous	Relative humidity of incoming air (%)
<code>inlet_temp_c</code>	Continuous	Temperature of incoming air (°C)
<code>vibration_mm_sec</code>	Continuous	Measured vibration velocity (millimeters/second)

3a. Fit a multiple linear regression model predicting `o2_output_lpm` using all five explanatory variables. Display the model results.

3b. State the model constructed.

$$\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x_1 + \cdots + \hat{\beta}_k x_k$$

3c. Construct a QQ plot of the residuals and a plot of residuals against fitted values. Do the graphs support the normality and constant variance assumptions? Explain.

Your answer here.

3d. Construct the overall hypothesis test to determine if at least one of the five explanatory variables has a linear association with oxygen output (i.e., is this a significant regression model?).

Hypotheses:

- H_0 :
- H_1 :

Test Statistic and p -value

- $statistic =$
- $p =$

Rejection Region

- Reject H_0 if ...

Conclusion and Interpretation

- Reject or fail to reject H_0 .
- There is or is not sufficient evidence to conclude...

3g. Frank says, “If the overall model is significant, every variable in it must matter.” Explain why that conclusion is not correct.

Your answer here.

3e. Run your model through `glm_table()` in the `ssstats` package. Then, identify which variables have a statistically significant linear association with oxygen output after accounting for the other variables. Do not include the intercept in this comparison.

```
# example for a model object named "m3"  
# m3 |> glm_table()
```

Your answer here.

3f. Interpret the coefficients for `airflow_lpm` and `filter_age_hours` in context. Include units and explain what is being held constant.

Your answer here - `airflow_lpm`

Your answer here - `filter_age_hours`

3g. Write two or three sentences to Mission Control summarizing what the model tells us about oxygen output.

Your answer here.